SAFL

Research Work Trial | Secure AI Futures Lab

Please read this doc carefully.

- This work trial should give you a good sense of research work at SAFL.
- We recommend a maximum of 12 hours of work, done inside a 48-hour window.
- We pay ₹1,000 per hour of work you record, up to a cap of ₹10,000.

Table of contents:

How the two days work 1
I. Scoping Note (~30 mins) 2
II. Reproduction (~4 hrs) 4
III. Extension (~5 hrs) 4
IV. Write-up and Executive Summary (~2 hrs) 5
V. AI Declaration and Worklog (~30 mins) 5
Using AI tools 5
Deliverables 6
Ground rules and tips 7

How the two days work

Your 48-hour window has now started. The end time must have been communicated to you in the email that shared this document.

1. **Work.** The window is 48 hours of clock time. We recommend no more than 12 hours of actual work within it, so you have enough time to step away. Treat it like a couple of actual days at work.
2. **Submission.** You share a single folder with us by the end of your window. See the deliverables section below for more info.

If something goes wrong during your window, email us at hiring@secureaifutureslab.com.

Time budget (Recommended)

Task	Recommended time
Problem statement reading (this document) + choosing the problem	30 mins
I. Scoping note	30 mins
II. Reproduction	4 hrs
III. Extension	5 hrs
IV. Write-up and executive summary	1.5 hrs
V. Worklog and AI declaration	30 mins
Total	**12 hrs**

You may move time between the sections as you see fit. Please state your actual time in your worklog. If you finish the work in less than the budget, that is fine.

I. Scoping Note (~30 mins)

Choose **one** problem from the set below. Then write **half a page** that covers:

- Which problem you chose, and why.
- The specific claim or question you are going to test.
- What you expect to find before you run anything.
- What result would change your mind.

Write this before you start coding, and do not edit it afterward. We will read it against your final write-up. If you predict wrong and notice that you predicted wrong, that is still a good outcome for us. If you quietly rewrite your hypothesis to match your results, we will see it.

The problem set

Choose one of the four problems below. We picked all four because a faithful reproduction is cheap (a laptop or a free-tier Colab GPU would be sufficient). We did this so that most of your time goes into the extension, and so that nobody is held back by the hardware they own.

1. **Option A. Chinchilla scaling, a replication attempt.** Besiroglu, Erdil, Barnett and You (2024), arXiv:2404.10102
    a. The authors reconstruct the data behind Hoffmann et al.'s compute-optimal scaling law.

   b. They argue that the third estimation procedure in the original paper fails to fit its own data and reports confidence intervals that are far too narrow for the number of experiments the authors ran.
2. **Option B. Adding error bars to evals.** Miller (2024), arXiv:2411.00640
   a. This paper treats the questions in an evaluation as a sample drawn from a larger set of questions that nobody ever sees in full.
   b. From that starting point, it derives standard errors, paired comparisons, cluster adjustments, ways to reduce variance, and power analysis for language model evaluations.
3. **Option C. Documenting large webtext corpora, a case study on C4.** Dodge, Sap, Marasović et al. (2021), arXiv:2104.08758
   a. An audit of a pretraining corpus that many people use.
   b. The authors ask where the documents came from, what the filters removed, how much machine-generated text survived, and how much of the text overlaps with benchmark test sets.

Please pick from these three. We ask everyone to work from the same problem set so that we can compare submissions fairly, so please do not substitute a problem of your own. If you have already worked on one of these four papers, say so in your scoping note and tell us what you did before the trial.

II. Reproduction (~4 hrs)

Reproduce one or more of the core results of the paper you chose.

Pick the result that you can extend in a non-trivial way, because the extension is worth more. Report the reproduction's results. If your number differs from the published one, report your number and give your best account of the gap. Do not round towards the paper.

Deliverable: working code, plus a short note on what matched, what did not, and what you assumed.

III. Extension (~5 hrs)

Take the result you reproduced and push on it, so that you produce knowledge the paper does not already contain. Once you have a hypothesis, ask yourself this question: **will this add to what is known, or teach me something I did not know, or produce a result that would surprise a reader?** If the answer to all three is no, pick a different extension.

A good extension usually does one of the following:

1. **Tests robustness.** Does the result survive a change in scale, dataset, model family, seed, or hyperparameter that the authors did not vary?
2. **Tests the construct.** Is the thing being measured the thing the authors think is being measured? Is there a simple baseline that explains the result just as well?
3. **Tests generalization.** Does the method still work on data or models released after the paper was written?
4. **Puts a number on an unstated assumption.** The paper asserts X. What happens across the range where X is false?

We will check four things in your design:

1. You stated your hypothesis before you ran the experiment.
2. You included a control or a baseline.
3. You quantified uncertainty. If you report a difference, tell us whether you can tell it apart from noise, and how you know. Stay close to the fundamentals rather than reaching for an unusual metric.
4. You gave at least one alternative explanation for your own finding, and said what evidence would separate it from your explanation.

Negative results are welcome. Done well, they score higher than a positive result with weak controls. A submission that says "I predicted X, I found no effect, and here is why I believe the null rather than believing my setup was broken" is a strong submission.

IV. Write-up and Executive Summary (~1 hr 30 mins)

Send us two documents alongside your code, plots, and other files. We weigh communication heavily. Unclear writing about good work reads the same to us as bad work.

1. **Executive summary, one page maximum.**
    a. What you set out to test, what you found, how confident you are, and what you would do next if you had another week.
    b. Write it assuming the reader stops after this page.
    c. If your executive summary is unclear and not well evidenced, we may not read the full write-up.
2. **Full write-up, five pages maximum, not counting appendices.**
    a. Treat this as a short research note written for your research lead.
    b. Cover the claim you looked into, your reproduction and any discrepancies, your hypothesis, your experimental design and why it is the right design, your results with their uncertainty, the alternative explanations you considered and could not rule out, and the limits of what you did.

2 hours is a sufficient time to spend on both documents.

V. AI Declaration and Worklog (~30 mins)

AI declaration. Add a short declaration at the end of each deliverable, following the policy in the next section.

Worklog. Keep a rough log of how you spent the time, in order. Write about 5 to 10 minutes of notes for every hour of work, as you go. It does not need to be polished, and we are not marking the prose.

Write the worklog yourself, in your own words. Do not draft it with an AI tool, and do not clean it up with one afterward. Rough notes in your own voice are good enough for us.

We want to see the shape of your process, so tell us where you got stuck, what you abandoned, and when you changed direction and why. Dead ends are useful to us, and we read this section carefully.

Using AI tools

We use AI tools in our own research every day, and we expect you to use them here. We are specific about where they help and where they do not.

Where we expect you to use them. Writing and debugging code, looking up methods, reading around the paper, and checking your own reasoning.

Where we ask you not to use them.

- Your scoping note. Write your own hypothesis, in your own words.
- Your worklog. See the section above.

You may draft the executive summary and the full write-up with AI help, but the reasoning and the words must end up as yours. We will reject a write-up that reads like unedited model output.

What to declare.

- A short statement at the end of each deliverable saying whether you used AI tools and how.
- Your transcripts, or links to them, in your worklog. Shareable chat links are the fastest way to do this, and we prefer them to screenshots.
- If you used an agentic coding tool such as Claude Code, Cursor, or Codex, say so and tell us roughly what fraction of the code it wrote.

You are responsible for your code running and your claims being true. A plot that a model produced and you cannot explain is worse than no plot.

Deliverables

1. Worklog document
2. Executive summary document
3. Full write-up
4. Code, plots, README, and any other files
5. AI usage declaration, transcripts, and chat links

Put everything into a single Google Drive folder and share the link with team@secureaifutureslab.

1. Give **team@secureaifutureslab.com** editor access and permission to copy the files.
2. Your code should run. Include a brief README that says how to run it.
3. If your work needs data that is too large to share, share the script that fetches it instead. You can host data on a free tier service such as Google Drive or Dropbox.

Compute and costs. All four problems can be completed on a personal laptop or a free tier Colab GPU. If your approach needs more than that, say so in your scoping note.

Ground rules and tips

1. **Do the obvious thing first.** Sophisticated methods are easy to reach for but also easy to waste hours on. If a scatter plot and a t-test answer your question, use them and spend the time you saved on thinking.
2. **Name your sources.** Cite your sources, datasets, and figures so that we can follow your chain of reasoning. If a number came from a paper, cite the paper. If it came from your own run, say so.
3. **Original framing counts.** The submissions we value most notice something we did not expect. A reproduction that comes back exactly as published and stops there is a pass, but not a strong pass.
4. **Reading is as important as producing.** Read the paper properly, and read at least a little of what has been written about it since. Several of these results are contested. Finding a rebuttal and engaging with it is a good signal.
5. **Scope ruthlessly.** The most common failure is trying to do too much and finishing nothing. We would rather see one question answered well than four questions opened.
6. **Report the failures.** This is what we look for most. Did you report something that did not work, and did you correctly diagnose why? Reproductions fail constantly.

7. **Keep it to yourself.** Please do not share this document or the problem set with anyone, and please do not post it publicly. We may use the same problems with other candidates.

Questions? Email us at hiring@secureaifutureslab.com